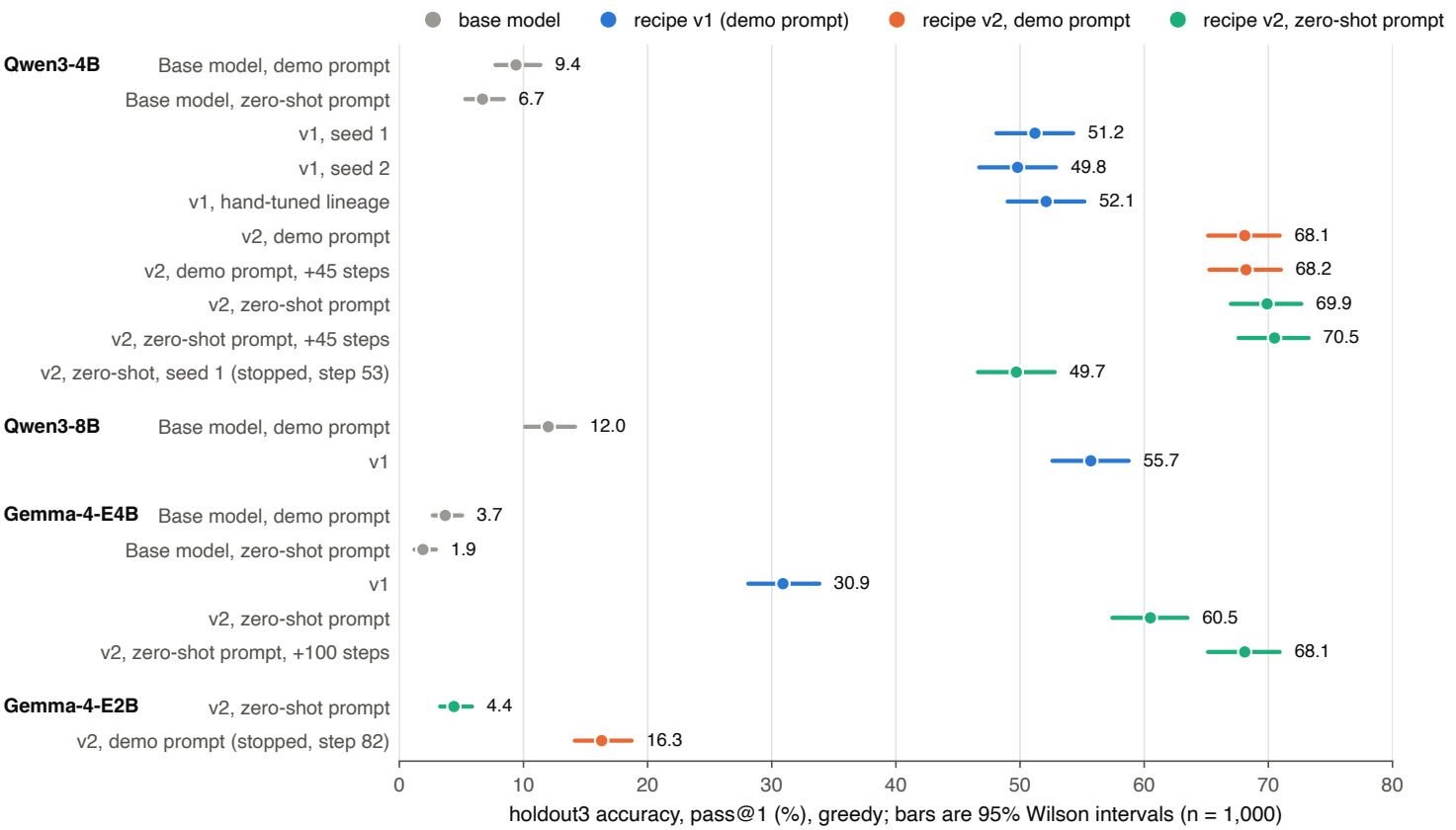
base model
recipe v1 (demo prompt)
recipe v2, demo prompt
recipe v2, zero-shot prompt
Qwen3-4B
Base model, demo prompt
9.4
Base model, zero-shot prompt
6.7
v1, seed 1
51.2
v1, seed 2
49.8
v1, hand-tuned lineage
52.1
v2, demo prompt
68.1
v2, demo prompt, +45 steps
68.2
v2, zero-shot prompt
69.9
v2, zero-shot prompt, +45 steps
70.5
v2, zero-shot, seed 1 (stopped, step 53)
49.7
Qwen3-8B
Base model, demo prompt
12.0
v1
55.7
Gemma-4-E4B
Base model, demo prompt
3.7
Base model, zero-shot prompt
1.9
v1
30.9
v2, zero-shot prompt
60.5
v2, zero-shot prompt, +100 steps
68.1
Gemma-4-E2B
v2, zero-shot prompt
4.4
v2, demo prompt (stopped, step 82)
16.3
0
10
20
30
40
50
60
70
80
holdout3 accuracy, pass@1 (%), greedy; bars are 95% Wilson intervals (n = 1,000)